Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

V SEMESTER B.TECH. (ELECTRICAL & ELECTRONICS ENGINEERING)

MAKEUP EXAMINATIONS, DECEMBER 2024

POWER SYSTEM ANALYSIS [ELE 3121]

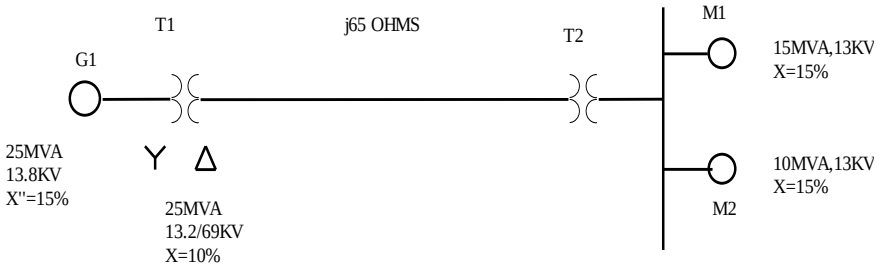
REVISED CREDIT SYSTEM

Time: 3 Hours **24 December 2024** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. Draw the per unit reactance diagram for the power system shown in fig Q1(A), choosing base values of 25MVA, 13.8kV in the generator circuit. The transformer T2 comprises of three single phase units each rated 10 MVA, 69/7.62 kV, X=10%. 4



FigQ1(A)

1B. Explain the tests to be conducted on a three winding transformer to obtain its reactances in per unit. Also draw the per unit reactance diagram. 3

1C. A 40 MVA, 22 kV synchronous generator has subtransient, transient and steady state reactances of 0.15 pu, 0.2 pu and 0.3 pu respectively based on its rating. For a symmetrical fault at its terminals on no load, determine 3

(i) Subtransient symmetrical fault current in Amp.
(ii) Sustained symmetrical fault current in Amp.
(ii) Momentary current of circuit breaker in Amp.

2A. A synchronous generator and a synchronous motor each rated 20MVA, 12.66kV having 15% reactance are connected through two transformers and a line. The transformers are rated 20MVA, 12.66/66KV and 66/12.66kV with leakage reactance of 10% each. The line has a reactance of 8% on base of 20MVA, 66 kV. The motor is drawing 10MW at 0.8 leading power factor and a terminal 4

voltage 11KV when symmetrical three-phase fault occurs at the motor terminals. Determine the fault current in per unit using Thevenin's theorem. Also determine the generator and motor currents in per unit. Choose base of 20MVA, 66 kV in the line.

2B. A 5MVA, 6.6kV 3-ph alternator was subjected to different types of faults for which the fault currents are as follows: 3-phase fault - 950A; Single line to ground fault - 1900A and Line to line fault - 1200A. Estimate the three-sequence component reactance of the alternator in per unit based on its rating. 3

2C. Derive an expression for the fault current when a L-L (line to line) fault occurs in a power system by clearly stating the terminal conditions. Also show the connection of the sequence diagrams for the above fault. 3

3A. A synchronous generator supplies power to two synchronous motors. The positive, negative and zero-sequence reactance of generator are 0.4, 0.3 and 0.1 pu respectively. The positive, negative and zero-sequence reactance of each motor are 0.3, 0.2 and 0.2 pu respectively. The neutral of generator is grounded through a reactance of 0.05 pu, one motor is solidly grounded while the other motor neutral is isolated. Find the fault current for a single line to ground fault at generator terminals. 4

3B. Consider a generator supplying power to an infinite bus through a transmission line. A 3-phase occurs at the terminals of the generator. Using equal area criterion, derive the expression for critical clearing angle. Draw relevant power-angle diagram showing accelerating and decelerating areas. 3

3C. A 50 Hz, Synchronous generator has H=5 MJ/MVA, and Xd'=0.3 p.u. It is connected to an infinite bus through a transformer and a double circuit line. The transformer reactance is 0.2 pu. Each line has a reactance of 0.3 pu. The voltage behind transient reactance of generator is 1.17 p.u. A three-phase fault occurs at the middle of one of the lines. The fault is cleared by the opening of the faulted line. Determine the critical clearing angle. 3

4A. Consider a 50 Hz generator delivering a real power of 0.9 pu to an infinite bus. The inertia constant H=2.7 s. The maximum power prior to the fault is 2.45 pu and during the fault is 0 pu. Choosing a time interval of 0.05 s compute the rotor angle for a sustained fault. Take the simulation time as 0.4 s. 3

4B. The rotor of an alternator is subjected to an acceleration of 15 electrical rad/s^2. If this acceleration exists constantly, compute the change in rotor angle and the speed in rpm at the end of 5 cycles. H = 5 MJ/MVA. Frequency = 50 Hz. Initially the machine is running at a normal speed without any acceleration. Number of poles = 2. 3

4C. The following is the system data for a load flow solution: 4

The line admittances:

Bus code	Admittance
1-2	2–j8.0
1–3	1–j4.0
2–3	0.666–j 2.664
2–4	1–j4.0
3–4	2–j8.0

The schedule of active and reactive powers:

Bus code	P	Q	V	Remarks
1	-	-	1.06	Slack
2	0.5	0.2	1+j0.0	PQ
3	0.4	0.3	1+j0.0	PQ
4	0.3	0.1	1+j0.0	PQ

Determine the voltages at bus number 2 and 3 the end of first iteration using Gauss-Seidel method. Take α = 1.6.

5A. Figure Q5 A shows the single line diagram of a simple three-bus system with generation at bus 1. The magnitude of voltages at bus 1 is adjusted to 1.04 per unit. The scheduled loads at bus 2 and 3 are given in the diagram. Build the bus admittance matrix [Ybus]. 3

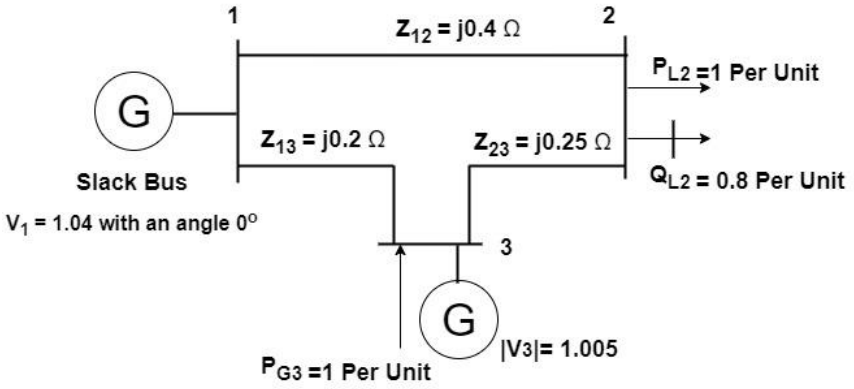


Fig Q5.A.

5B. Develop a diagonal elements of Jacobian matrix with the consideration of data given in Question 5A and figure Q5.A. 4

5C. Describe the step-by-step computation procedure of load flow solution by Newton-Raphson method using polar coordinates. 3